SAIM ZAFAR

Full-Stack Developer | Machine Learning | BSIT, Bahria University Islamabad

0311-5397891 | saim120zafar@gmail.com | github.com/SaimZafar | linkedin.com/in/saimzafar

PROFESSIONAL SUMMARY

Full-Stack Developer and machine-learning practitioner with a strong academic record (CGPA 3.90) and hands-on experience building and deploying end-to-end web and ML applications. Skilled in React, Node.js/Express, FastAPI, REST APIs, MongoDB and MySQL, with secure authentication (JWT, OAuth) and cloud deployment via Docker, Vercel, Render, and Hugging Face Spaces. Experienced across full-stack development, deep learning (PyTorch), and AI integration, focused on building secure, scalable, and explainable systems.

EDUCATION

Bachelor of Science in Information Technology (BSIT)
Bahria University, Islamabad | CGPA: 3.90 / 4.00 | Merit scholarship every eligible semester (GPAs 4.0, 3.91, 3.94)

A-Levels, Computer Science
Asas International School, Islamabad | Aug 2021 - Jan 2023 (Mathematics A, Physics B, Computer Science C)

IELTS Academic | Overall Band 7.0 (May 2023)

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, C, SQL
Frontend: React, HTML5, CSS3, JavaScript, jQuery, AJAX
Backend: Node.js, Express.js, FastAPI, REST API Design, JWT, OAuth
Databases: MongoDB, MySQL, Oracle
Machine Learning / Data: PyTorch, scikit-learn, NumPy, Pandas, Matplotlib, CNNs, Grad-CAM
DevOps & Cloud: Docker, Docker Compose, Git/GitHub, Vercel, Render, Hugging Face Spaces, Supabase
Testing & Other: Jest, Cisco Packet Tracer, Qt

PROJECTS

ECG-Scope - Explainable ECG Diagnosis Platform (Deep Learning, Full-Stack) Live Demo | GitHub
Tech: Python, PyTorch, FastAPI, React, Supabase, Docker, scikit-learn, Hugging Face Spaces, Vercel
- Built and deployed a full-stack platform that classifies 12-lead ECGs into five diagnostic categories and produces Grad-CAM visual explanations of each prediction
- Trained a 1D convolutional neural network from scratch on the PTB-XL dataset (~21,000 ECGs), reaching 0.92 macro-AUC with patient-disjoint splits and 3-seed validation
- Ran a cross-dataset generalization study across three hospital datasets; diagnosed BatchNorm covariate shift and applied label-free test-time adaptation (AdaBN) to recover performance to 0.87 macro-AUC
- Served the model through a FastAPI REST API (Dockerized on Hugging Face Spaces) and a React frontend (Vercel) with Supabase authentication and per-user analysis history

ModGuard - AI Content Moderation Platform (Full-Stack, AI Integration) GitHub
Tech: Node.js, Express.js, MongoDB, Mongoose, React, JWT, Google Gemini API, Multer, Docker
- Built a full-stack platform that screens user-submitted images with the Google Gemini API across six policy categories, returning Approved / Flagged / Blocked verdicts with per-category confidence and reasoning
- Designed a policy-driven verdict engine with admin-configurable confidence thresholds and enforcement rules, and snapshotted the active policy per submission for a reproducible audit trail
- Implemented JWT authentication with role-based middleware, a user appeals workflow with admin override, and a Recharts analytics dashboard (volume, verdict distribution, appeal outcomes)
- Abstracted the AI provider behind a single service with safe-by-default fallback; containerized MongoDB, the API, and the frontend with Docker Compose

Nexora - Full-Stack Online Marketplace (Team Project) GitHub
Tech: Node.js, Express.js, MySQL, Sequelize ORM, AWS S3, JWT, Passport.js, jQuery, AJAX, Vercel, Render
- Architected and built RESTful APIs for user management, product listings, cart, checkout, and order history
- Implemented multi-strategy authentication: Google OAuth, Facebook OAuth, and local login using Passport.js and JWT
- Integrated AWS S3 for scalable product image storage; secured the API layer with bcrypt password hashing and Helmet
- Deployed a cloud-native architecture: frontend on Vercel, backend on Render, database on Clever Cloud MySQL
- Added an admin panel, CSV data export, QR-code generation for orders, and real-time AJAX-based UI interactions

SkillLink - Freelance Marketplace System [GitHub]

Tech: Oracle XE 21c, Node.js, Express.js, React.js, REST API, JWT, Role-Based Access Control

- Built a full-stack freelance marketplace with 15 fully normalized Oracle tables covering registration, project posting, competitive bidding, contracts, milestones, and reviews
- Designed an ER model using disjoint-total specialization, weak entities, and junction tables; all tables satisfy 3NF with referential integrity via primary, foreign, and check constraints
- Exposed 35 REST API endpoints with three multi-table transactions and role-based access control for clients, freelancers, and admins
- Runner-Up, IEEE Department Innovation Challenge - Semester Project Showcase (Spring 2026), Bahria University

NovaCORE - OS Concepts Simulator with xv6 Kernel Extension [GitHub]

Tech: C (C99), POSIX pthreads, GCC, QEMU, xv6 Kernel

- Built a terminal-based OS simulator in C covering six modules: Round Robin/SJF/Priority scheduling, FIFO/LRU/Optimal page replacement, Producer-Consumer IPC, deadlock detection (Banker's Algorithm), and process state simulation
- Added a real getmeminfo() system call (syscall #22) to the xv6 teaching kernel by modifying nine kernel source files for live physical-memory reporting inside QEMU
- Compiled cleanly with zero warnings (GCC 13, -Wall -std=c99); diagnosed and fixed a Round Robin clock bug, a semaphore race condition, and GCC 13 compatibility issues

Competria - Competitive Programming Platform [GitHub]

Tech: C++, Qt Framework, QScintilla, OOP

- Built a desktop problem-solving platform with role-based access control, secure authentication, and admin dashboards
- Implemented dynamic problem creation, code submission, and automated evaluation with real-time feedback
- Integrated the QScintilla code editor for in-app coding; designed a modular OOP architecture for scalability

Python Data Science & ML Implementations [GitHub]

Tech: Python, NumPy, Pandas, scikit-learn, Matplotlib

- Implemented KNN, Decision Trees (ID3), Random Forest, SVM, and PCA from scratch, then benchmarked against scikit-learn
- Performed an end-to-end ML workflow: data preprocessing, visualization, model training, hyperparameter tuning, and evaluation

Enterprise Network Design & Simulation (Team Project) [GitHub]

Tech: Cisco Packet Tracer, VLANs, DHCP, RIP v2, TFTP, CIDR Subnetting

- Designed a full enterprise network topology with 3 routers, 4 switches, and 4 departmental VLANs (HR, Admin, Finance, IT)
- Configured dynamic IP assignment via DHCP, inter-department routing with RIP v2, and router backup via a TFTP server
- Applied CIDR-based subnetting and security configurations including password protection and hostname management

ACHIEVEMENTS & CERTIFICATIONS

- Merit-Based Scholarship (1st Position) - awarded in the 1st, 2nd, and 4th semesters at Bahria University
- Rector's Honor List - 1st, 2nd, and 4th semesters
- 5th Place - CODEZAAR Competitive Programming Competition, Bahria University (multi-university event)
- Runner-Up (IEEE Award) - Department Innovation Challenge, Semester Project Showcase, Bahria University Islamabad (Spring 2026)
- Python for Data Science - Udemy
- Participant - Youth Speak Forum 2025 (AIESEC Islamabad)